TrialFair — Investigating sex-specific safety gaps in clinical trials

TrialFair helps pharmaceutical researchers identify what a clinical trial's safety results reveal about women—and what remains unknown—before designing the next study.

The problem and who it helps. Underrepresentation of women, particularly older women, can leave gaps in treatment safety evidence (Daitch et al., 2022), while sex differences in drug exposure and elimination can contribute to more adverse reactions in women (Zucker & Prendergast, 2020). The GAO found that eight of ten drugs withdrawn from the U.S. market between 1997 and 2000 posed greater health risks to women (GAO, 2001). Investigating these risks requires time-consuming comparison of trial results with fragmented scientific literature. TrialFair supports clinical safety scientists and trial designers reviewing completed studies, particularly at the Phase I → Phase II transition.

Our Solution. TrialFair combines participant representation, adverse events, drug-exposure measurements and published evidence through connected AI agents and deterministic statistical screening in Python. The Trial Data Interpreter interprets symptoms and adverse-event terminology. The Safety Investigator prioritises findings, develops explicitly labelled hypotheses and requests further analyses. The External Evidence Investigator evaluates supporting or contradictory publications. A fourth agent, in development, will propose follow-up analyses and improvements to subsequent trials. Agents share structured results and source references; a dashboard presents findings, limitations and recorded agent activity.

Our data. Our demonstration uses three clearly labelled synthetic participant-level datasets representing fictional hypnotic, antiseizure and cardiac-repolarisation drugs. They contain simulated demographics, adverse events, participant-level pharmacokinetic parameters and narratives, with ECG (QTcF) measurements for the cardiac dataset. Each generator is calibrated to published findings and product labels across several approved drugs in its class, combined with stated illustrative assumptions, so no single real drug is a direct match. Sex differences arise from a simulated mechanism (drug exposure, baseline QTc or body weight) rather than fixed rates, and each dataset's known truth is withheld from the agents for evaluation. These are simulated observations, not real patient records.

For demo speed, each synthetic dataset is paired with two preselected real publication excerpts and prepared annotations. These provide drug-class context, not direct evidence about the fictional compound. Separately, the application supports ClinicalTrials.gov records and budget-limited Europe PMC literature searches; an alternative demo uses a captured ACTT-1 record. Missing data remain explicit, and source records retain content hashes, timestamps, access levels and licence metadata where available.

How we evaluated it. Evaluation covered software reliability, statistical behaviour and evidence grounding. The implementation passed 314 offline tests covering calculations, cross-agent consistency, citations, contradictory evidence and failure handling. Known-truth synthetic trials measured detection and false-positive rates for selected statistical tools across simulated mechanisms and negative controls; they do not establish real-world or agent-level accuracy. A live literature check produced two accepted evidence items with quotations matched to retrieved abstracts, but one contextual formulation comparison remained unverified. The integrated demo preserved citations and uncertainty, using labelled

fallbacks when final model outputs failed validation. Reliable autonomous completion, clinical usefulness and researcher time savings remain unestablished.

Limits and safeguards. Python calculates statistics; model interpretations do not create participant records or event counts. Source, passage, quotation and reference checks establish evidence traceability, not scientific validity. Missing data, ambiguity and conflicting evidence remain explicit. Time and tool budgets bound execution; completed outputs are retained where possible and incomplete runs labelled. Models are instructed against unsupported causal or safety conclusions, and consequential recommendations require expert review. TrialFair proposes research actions; it does not prescribe patient-specific doses, approve trial progression or execute recommendations.

TrialFair's wider impact. TrialFair supports the hackathon's 'fairer' objective by making sex-specific safety evidence a clearer part of drug-development decisions. Prioritising follow-up investigations before larger trials could also make development more efficient and less costly. Stronger early-stage evidence could reduce the likelihood of safety issues requiring major investigation after approval, when corrective action may involve additional studies, revised prescribing guidance or withdrawal. These are intended benefits, not clinically demonstrated outcomes.

REFERENCES:

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